

Assignment Code: DS-AG-032

LLMs and Langchain Framework | Assignment

Instructions: Carefully read each question. Use Google Docs, Microsoft Word, or a similar tool to create a document where you type out each question along with its answer. Save the document as a PDF, and then upload it to the LMS. Please do not zip or archive the files before uploading them. Each question carries 20 marks.

Total Marks: 200

Question 1: What are Large Language Models (LLMs) and how do they function?

Answer:

Question 2: Discuss the impact of LLMs on traditional software development approaches.

Answer:

Question 3: What are the key advantages and limitations of using LLMs in real-world applications?

Answer:

Question 4: Describe how different industries are being transformed by the use of LLMs. Provide examples.

Answer:

Question 5: Compare and contrast Langchain and LamaIndex. What unique problems does each solve?

Answer:

Question 6: Implement a basic Langchain pipeline using OpenAI's LLM to answer questions based on a user input prompt.

(Include your Python code and output in the code box below.)

Answer:

Question 7: Integrate Langchain with a third-party API (e.g., weather, news) and show how responses can be generated via LLMs.

(Include your Python code and output in the code box below.)

Answer:

Question 8: Create a LlamaIndex implementation that indexes a local text file and retrieves answers from it.

(Include your Python code and output in the code box below.)

Answer:

Question 9: Demonstrate combining Langchain with LlamaIndex to create a simple document-based Q&A chatbot.

(Include your Python code and output in the code box below.)

Answer:

Question 10: A legal firm wants to use AI to summarize large volumes of legal documents and retrieve relevant information quickly. Propose a solution using Langchain and LlamaIndex, and explain how it would work in practice.

(Include your Python code and output in the code box below.)

Answer: